# Supplementary Material
# An Interpretable Statistical Surrogate of Activated Sludge Systems that Preserves Mass Conservation and Component Non-Negativity

## Target-Wise Interpretability Atlases

This supplementary file collects the complete target-wise ICSOR interpretability atlases that are not retained in the main manuscript. Figure 4 of the manuscript is now reserved for the COD $\Theta_{cc}$ block only, so the paper can discuss one representative interaction surface without repeating four dense multi-block panels in the main text. Figures S1–S4 provide the full COD, TN, TP, and TSS atlases.

Each atlas presents the full set of retained blocks for one reported target: the scalar bias term $b$, the first-order operational block $W_u$, the first-order influent block $W_{in}$, the operational-curvature block $\Theta_{uu}$, the operation-influent interaction block $\Theta_{uc}$, the influent-curvature block $\Theta_{cc}$, and the coupling matrix $\Gamma$. The ASM component basis is kept complete in every panel, and the heatmaps are drawn from the lower origin so the operational axes begin at HRT and then Aeration.

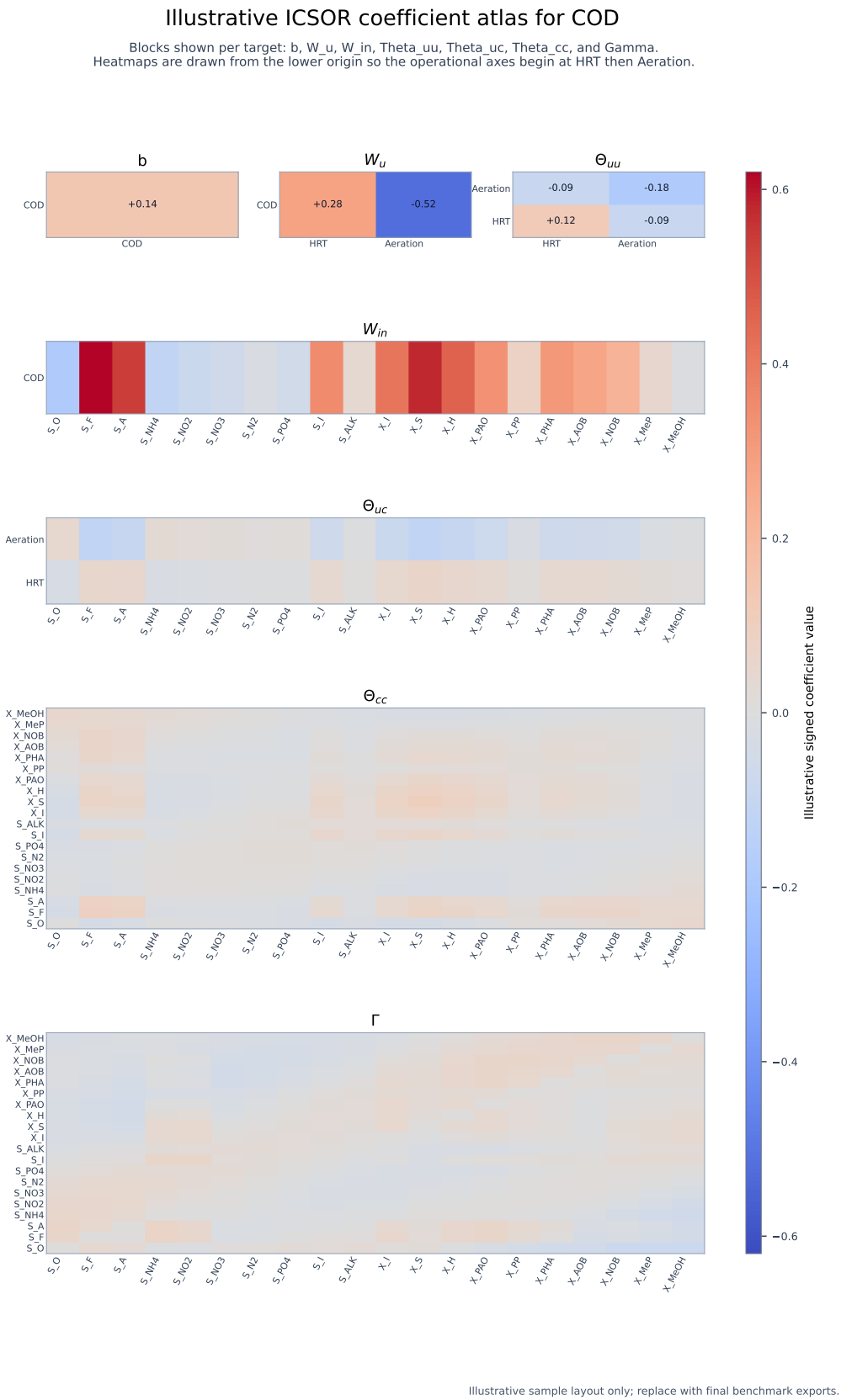


Figure S1: Illustrative COD-specific ICSOR coefficient atlas showing $b$, $W_u$, $W_{in}$, $\Theta_{uu}$, $\Theta_{uc}$, $\Theta_{cc}$, and $\Gamma$. Replace this sample asset with the final COD atlas selected for the supplementary narrative.

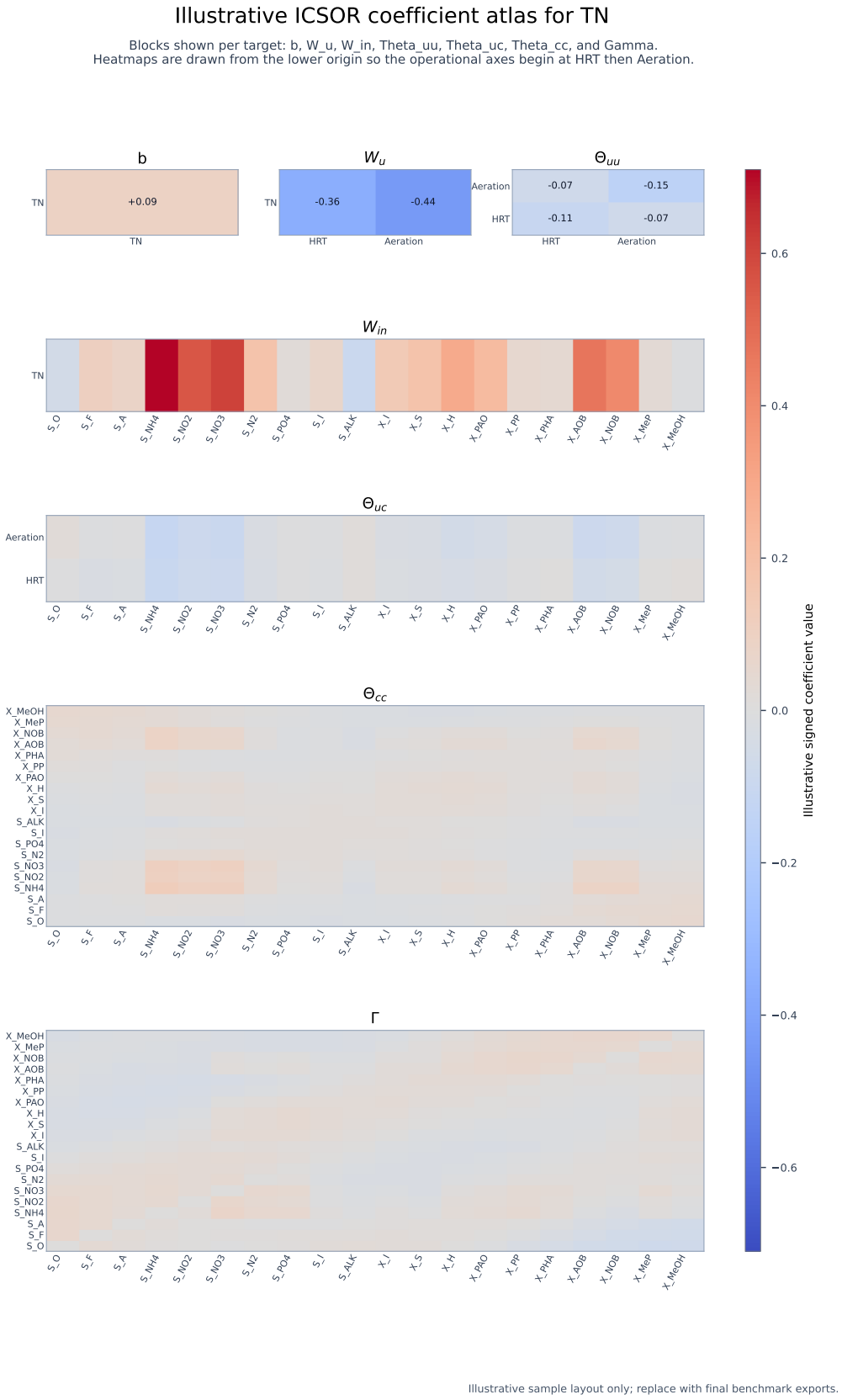


Figure S2: Illustrative TN-specific ICSOR coefficient atlas showing $b$, $W_u$, $W_{in}$, $\Theta_{uu}$, $\Theta_{uc}$, $\Theta_{cc}$, and $\Gamma$. Replace this sample asset with the final TN atlas selected for the supplementary narrative.

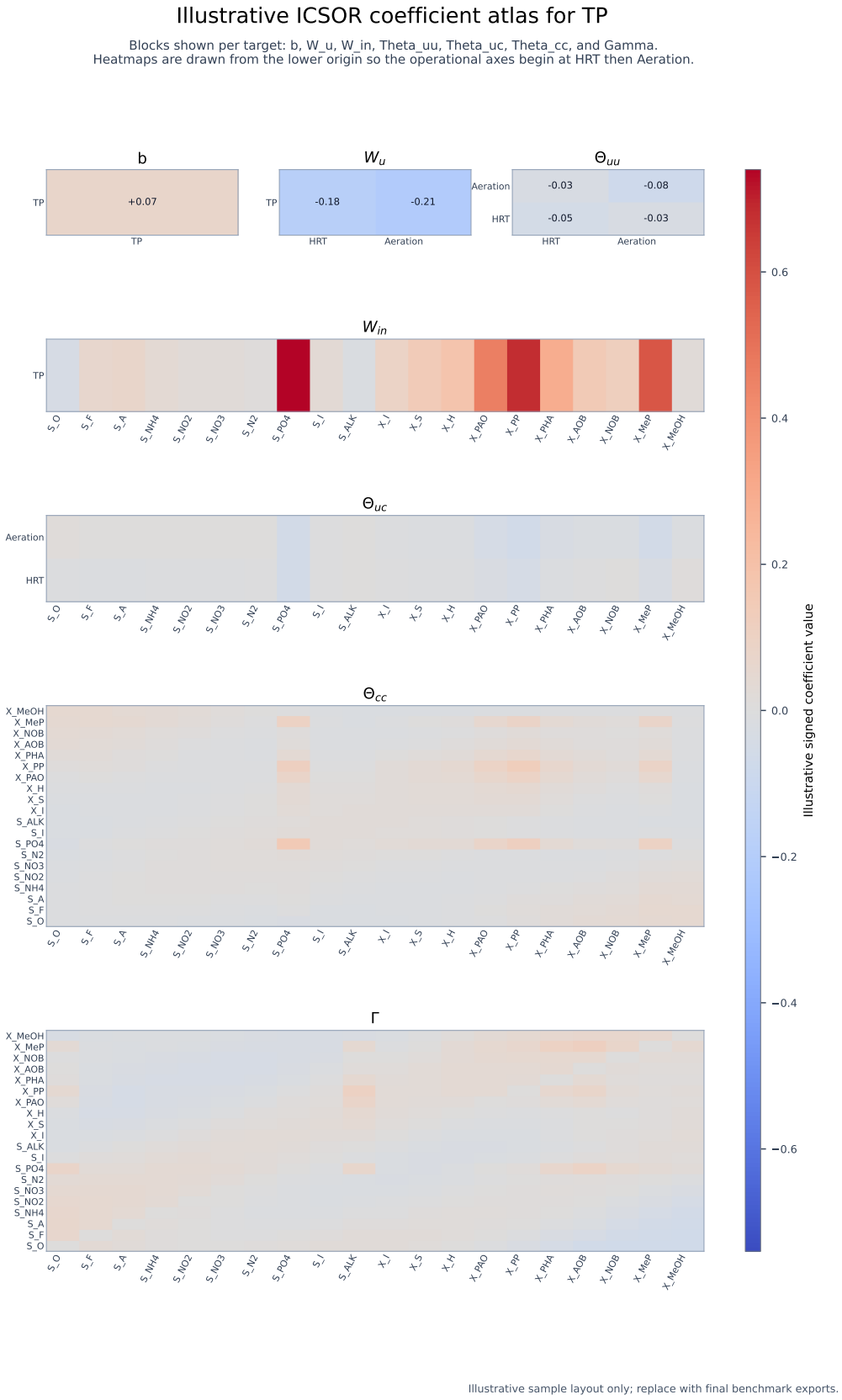


Figure S3: Illustrative TP-specific ICSOR coefficient atlas showing $b$, $W_u$, $W_{in}$, $\Theta_{uu}$, $\Theta_{uc}$, $\Theta_{cc}$, and $\Gamma$. Replace this sample asset with the final TP atlas selected for the supplementary narrative.

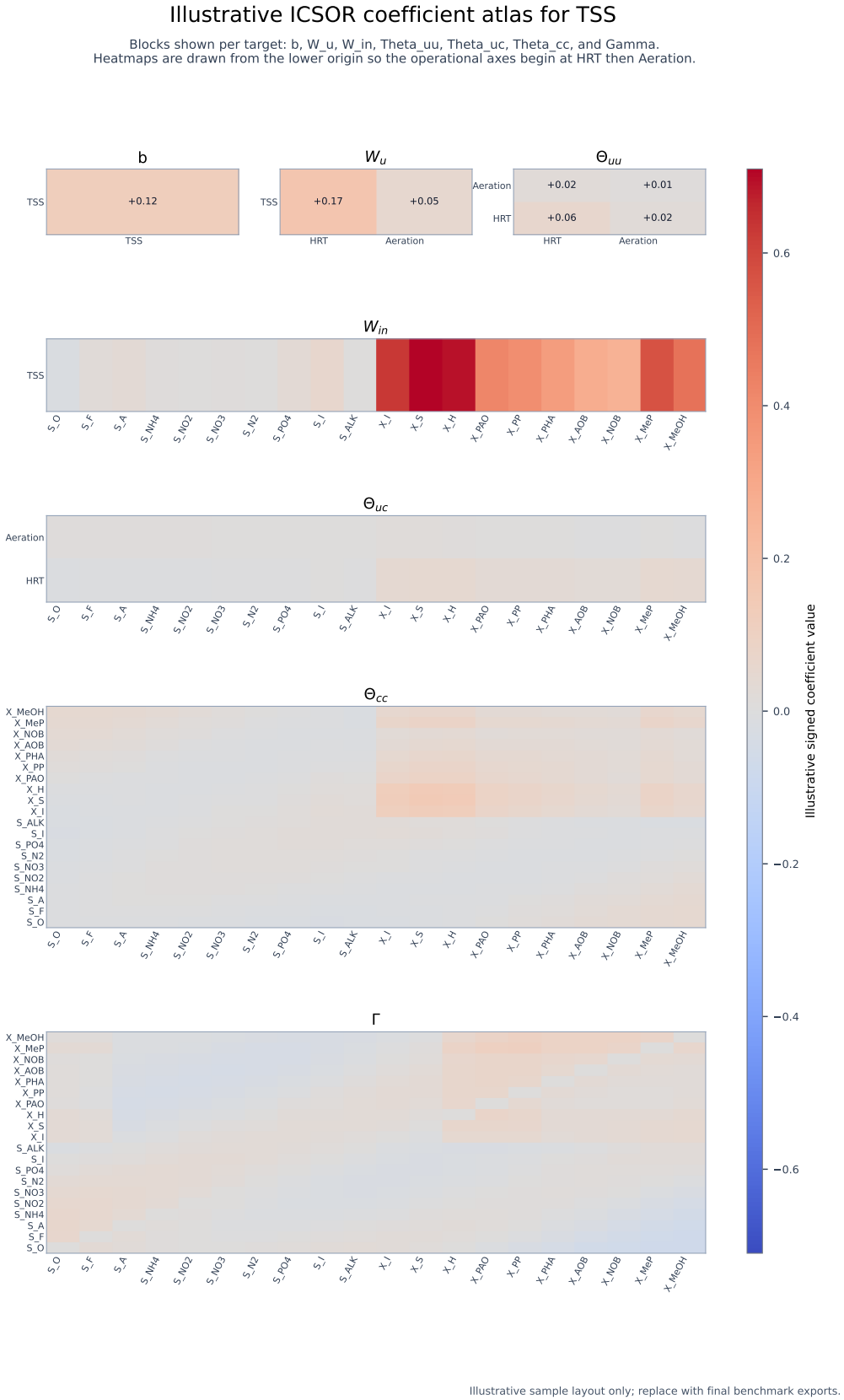


Figure S4: Illustrative TSS-specific ICSOR coefficient atlas showing $b$, $W_u$, $W_{in}$, $\Theta_{uu}$, $\Theta_{uc}$, $\Theta_{cc}$, and $\Gamma$. Replace this sample asset with the final TSS atlas selected for the supplementary narrative.